Auctions; Asymmetric Information

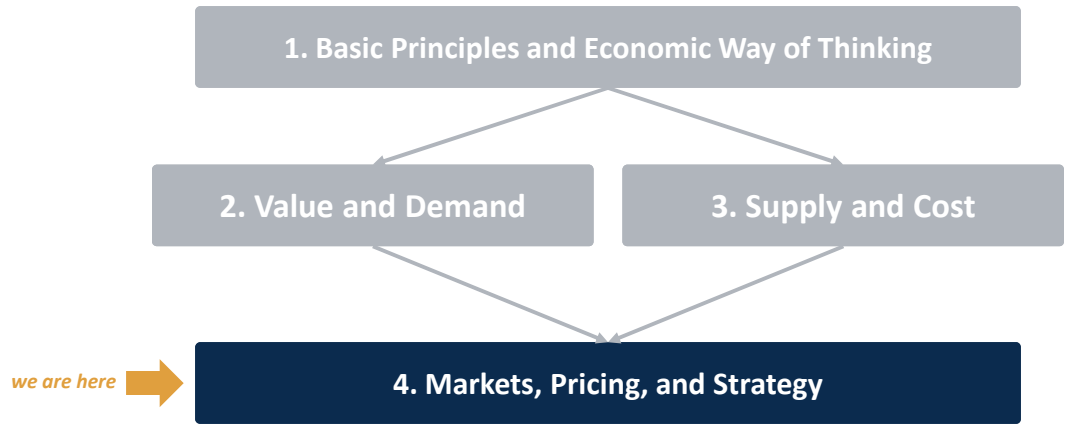
Module 8

Management 405

Prof. Nico Voigtländer · UCLA Anderson

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Agenda for the Class



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Module 8 · Agenda

Outline of Module 8

1	Auctions	In class
2	Asymmetric Information [if we have time – not in the Final Exam]	In class
2a	Principal-Agent Problem	In class
2b	Moral Hazard	In class
2c	Adverse Selection	In class
2d	Signaling	In class
3	Elective Classes in Economics	In class

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Module 8 · Agenda

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Auctions

- Auctions occupy a middle ground between markets with listed prices and individual deals / bargaining
- Auctions are an efficient way to find a buyer with high willingness to pay for items that are special or unique



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Extravagant Edibles: A $1.8 Million Bluefin Tuna, $7,500 Royal Wedding Cake Slice, and More



Kristin Tablang Forbes Staff
ForbesLife
I am the lifestyle assistant editor at Forbes.

This story appears in the February 28, 2017 issue of Forbes. Subscribe



Kiyoshi Kimura, president of sushi restaurant chain Sushi-Zanmai, poses with a 212-kilogram bluefin... [+]

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Advantages of Auctions

- **Better surplus extraction**, when compared to list prices
- **Speed and flexibility**, when compared to bargaining



Gemini Notebook

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Private Value vs. Common Value

Private Values	Common Values
▪ **Private values:** buyers differ in how much they value the items being purchased – Art auction: art lovers may have different willingness to pay for the same painting	▪ **Common values:** buyers agree on the item's value, but often the true value is unknown – E.g., auctioning off an oil field: everyone is uncertain about how much oil there is in the ground. If it were known, all would have the same valuation

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Storage Wars: Private Value or Common Value?



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Types of Auctions

The following are four well-known auctions:

Live	*Sealed bid*
English Auction	**First-Price Sealed Bid Auction**
Dutch Auction	**Second-Price Sealed Bid Auction**

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Live Auctions

English Auction

Bidders publicly announce bids in ascending fashion. *E.g., Sotheby's*

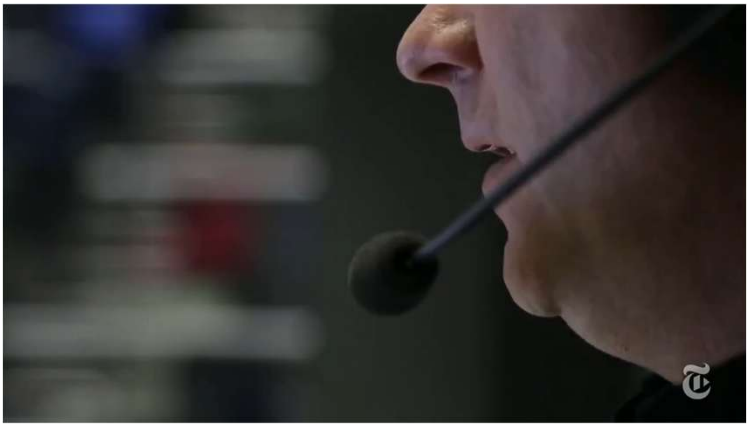


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Live Auctions

Dutch Auction

Descending auction: the auctioneer announces a high price and lowers it until someone decides to purchase at the going price. *E.g., Aalsmeer flowers*

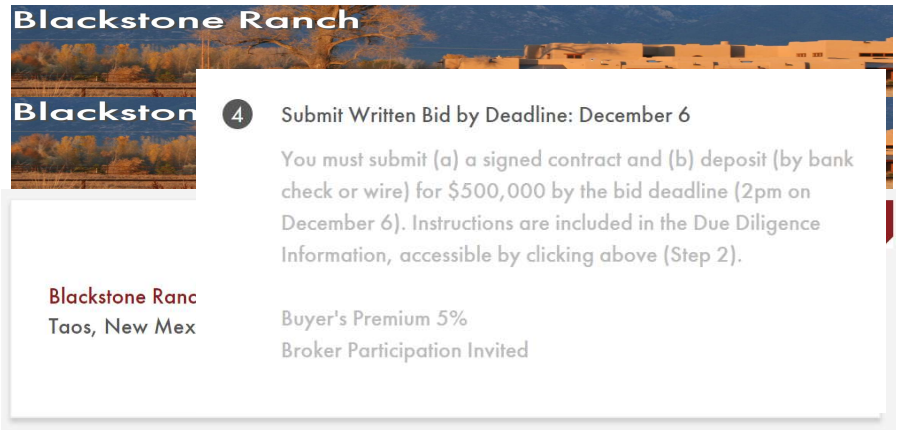


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Sealed-Bid Auctions

First-Price Sealed Bid Auction

Bidders submit sealed bids, not knowing what other players bid. The best bid wins. *E.g., buying a house*



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Sealed-Bid Auctions

Second-Price Sealed Bid Auction

Bidders submit sealed bids. The best bid wins, but the winner pays the amount bid by the second-best bidder. *E.g., Google AdWords*



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Optimal Bidding Strategies

English Auction (bidders publicly announce bids in ascending fashion)	Keep bidding up to your private value $b_i = v_i$, or slightly below. Final price is approximately $v_{2\text{nd}}$.
Second-Price Sealed Bid Auction (sealed bids, best bid wins, but pays second-best bid)	Bid **exactly** your private value, $b_i = v_i$. Final price is $v_{2\text{nd}}$.
First-Price Sealed Bid Auction (sealed bids, best bid wins)	Bid a little less than your true valuation, v_i. This ensures that you have *some* surplus if you win.
Dutch Auction (the auctioneer announces a high price and lowers it until someone decides to purchase)	Bid a little less than your true valuation, v_i. That is, wait to bid until the price falls sufficiently below v_i.

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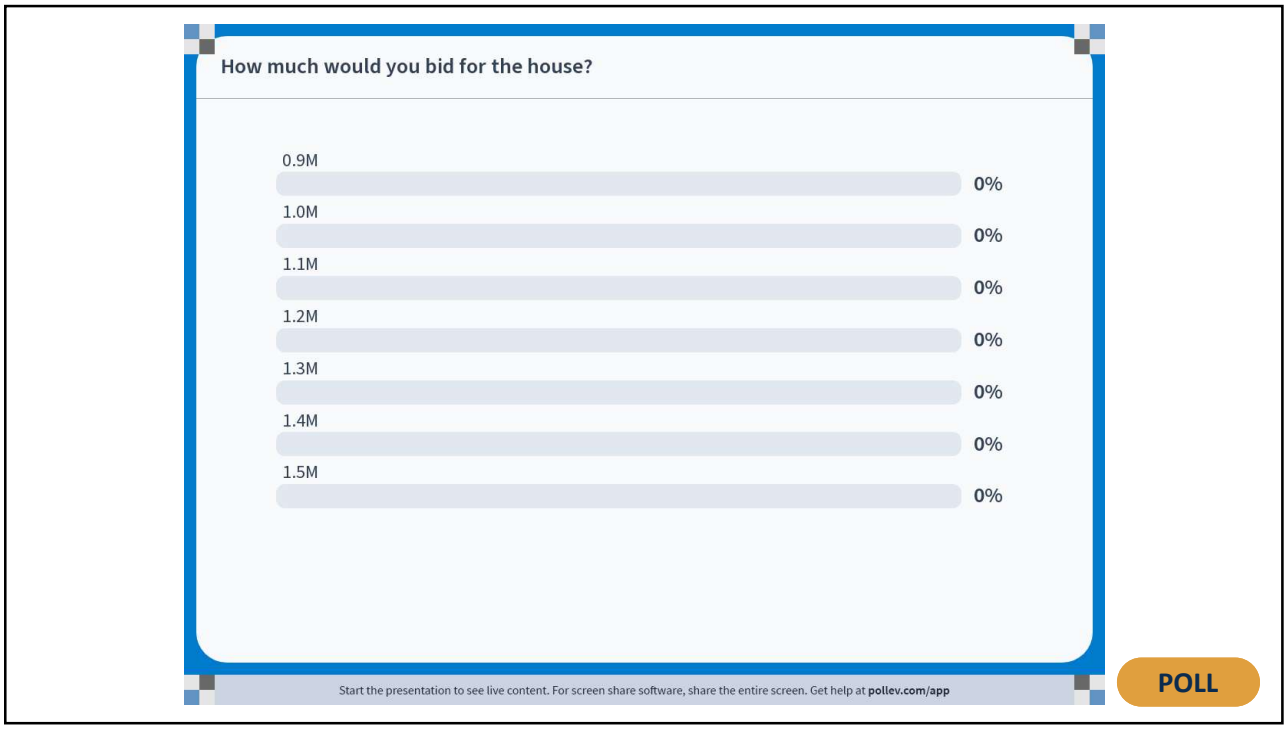
Bidding for a House



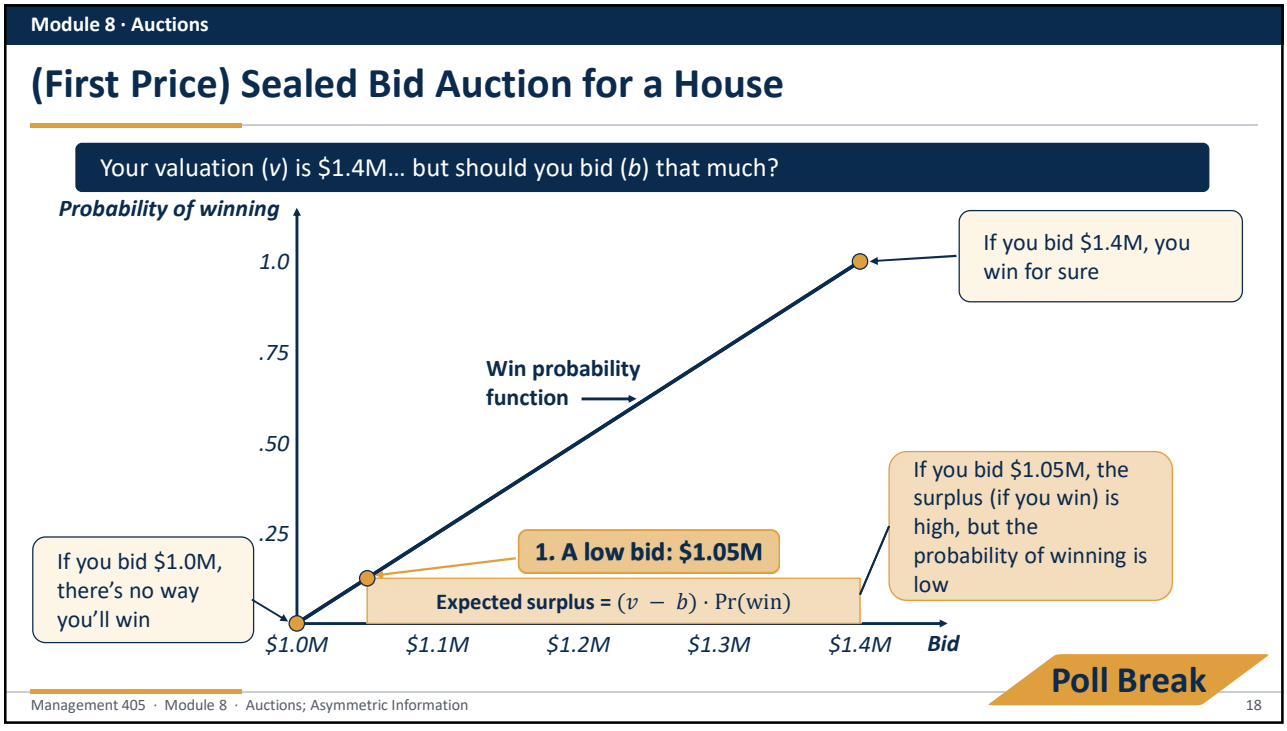
- You bid in a first-price sealed-bid auction for a house
- You value the house at $1.4M, but you'd like to get a good price on it
- You are bidding against one other bidder. Their valuation is between $1.0M and $1.4M
- **How much would you bid?**

Poll Break

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How much would you bid for the house?
0.9M
0%
1.0M
0%
1.1M
0%
1.2M
0%
1.3M
0%
1.4M
0%
1.5M
0%
Start the presentation to see live content. For screen share software, share the entire screen. Get help at pollev.com/app
POLL

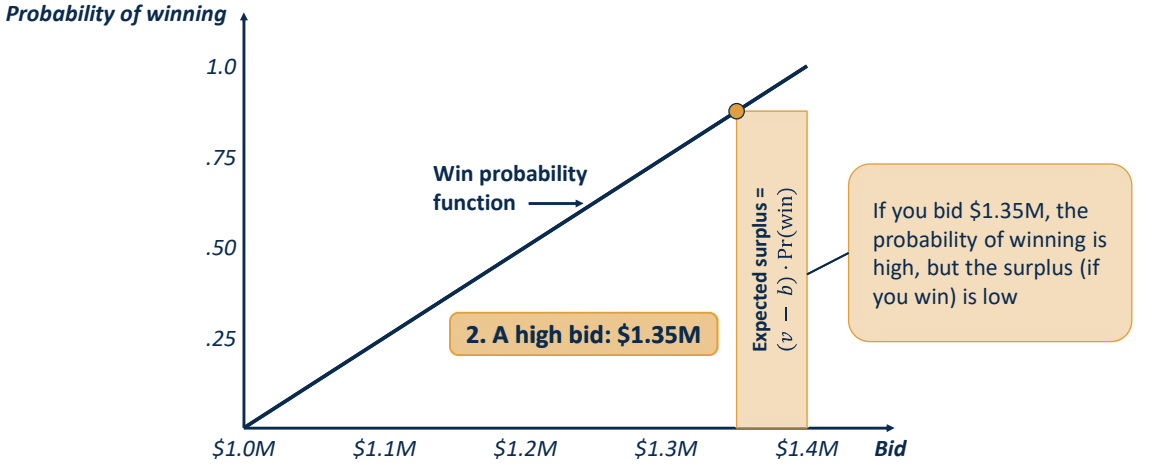
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Module 8 · Auctions
(First Price) Sealed Bid Auction for a House
Your valuation (v) is $1.4M… but should you bid (b) that much?
Probability of winning
1.0
.75
.50
.25
Win probability function
If you bid $1.4M, you win for sure
If you bid $1.05M, the surplus (if you win) is high, but the probability of winning is low
1. A low bid: $1.05M
Expected surplus = $(v - b) \cdot \Pr(\text{win})$
If you bid $1.0M, there's no way you'll win
$1.0M
$1.1M
$1.2M
$1.3M
$1.4M
Bid
Poll Break
Management 405 · Module 8 · Auctions; Asymmetric Information
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(First Price) Sealed Bid Auction for a House

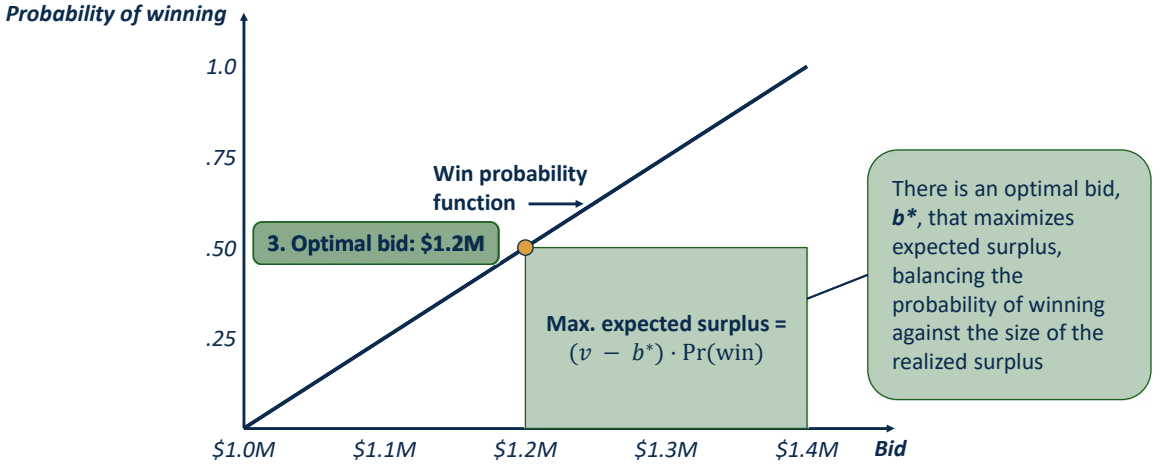
Your valuation (v) is \$1.4M



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(First Price) Sealed Bid Auction for a House

Your valuation (v) is \$1.4M



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Common Value Auctions

- Examples:
  - Auctioning off an oil field
  - FCC spectrum licenses for radio, cell phone, TV, etc.
  - Auctioning off a jar of pennies
- All bidders would agree on the value of the good, **if only it were known**

Jar of Pennies Auction

- Anyone care to bid **real money** for the contents of the jar?
  - 10 bidders
- There are $$X$ in the jar
- Rules:
  - If you bid $b < X$ and **win**, you get the difference ($X - b$)
  - If you bid $b > X$, you pay me the difference ($b - X$)

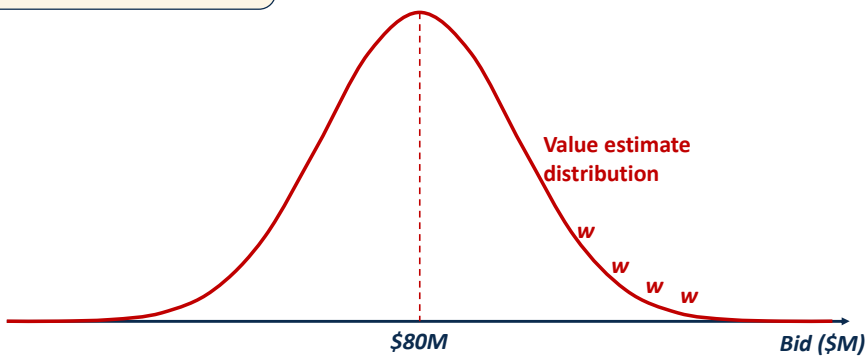


The Winner's Curse: Setup

- Suppose that you consider bidding on an **oil field**
- The auction is a **first-price sealed bid auction**
- Suppose the true value is **$80 million**, but it is unknown
- It is reasonable to assume that the **average bid** among many bidders will be $80 million
- By definition, the winning bid will tend to be on the high end of the distribution of bids
- It is possible, even likely, that the winner will bid more than the true value of the good: the **winner's curse**

The Winner's Curse: Illustration

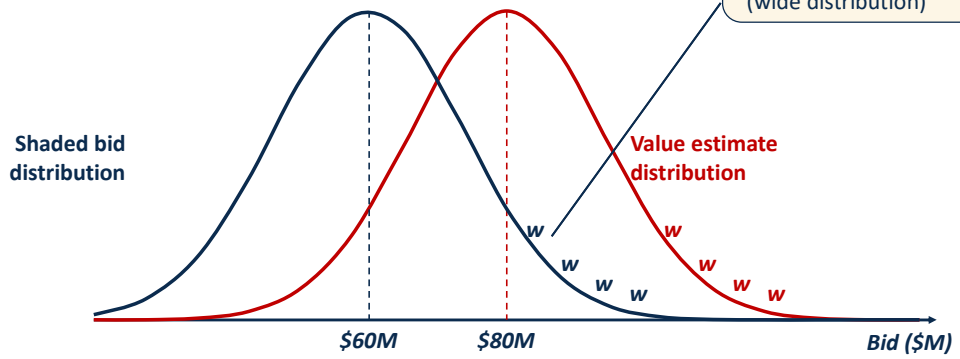
When the value of the auctioned item is highly uncertain, a winning bid (drawn from the right tail) will exceed the true value of the item



The Winner's Curse: Shade Your Bid!

All bidders discount bids below their estimates (by $20M in this example)

Nonetheless, winning bids may still exceed the true value of the item, especially under high uncertainty about the true value (wide distribution)



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The Winner's Curse: Remedies

- **Common value** auction: **shade your bid**
- The more bidders, the more you shade
  - If you win against 50 bidders, you likely **overestimated** the value
- **Private value** auction (sealed bid): shade **less** with more bidders
  - Shading too much makes you lose, even with the highest valuation

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Module 8 · Agenda

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1. Auctions — In class
2. Asymmetric Information [not relevant for the exam] — In class
   - 2a Principal-Agent Problem — In class
   - 2b Moral Hazard — In class
   - 2c Adverse Selection — In class
   - 2d Signaling — In class
3. Elective Classes in Economics — In class

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Module 8 · Principal-Agent Problem

Principal-Agent Problem

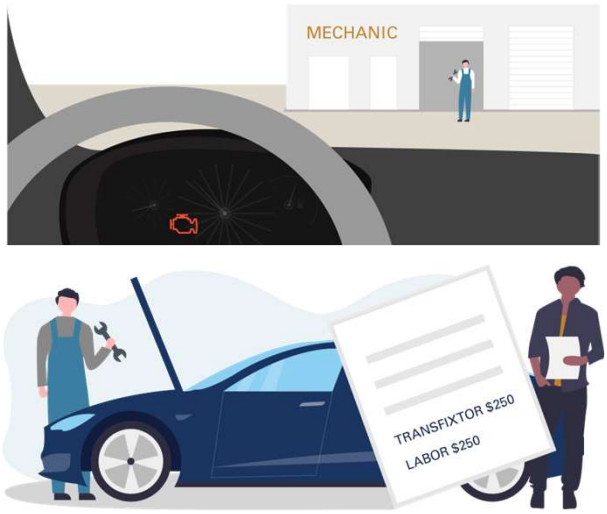
- Two parties:
  - **Agent**: person who acts (or recommends the action)
  - **Principal**: person affected by the action
- In a transaction between two individuals, one may have more information than the other:
  - **Asymmetric information**
- **Moral hazard** arises if:
  - Agent has private information
  - Agent and principal's preferences are misaligned



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Principal-Agent Relationship: Car Owner and Mechanic

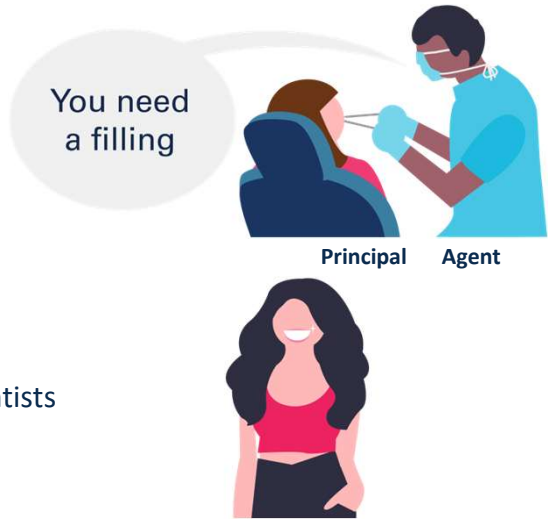
- **Principal:** Car owner
- **Agent:** Mechanic
- Asymmetric information
  - Mechanic knows more about the true problem
- Moral hazard
  - Mechanic may exaggerate



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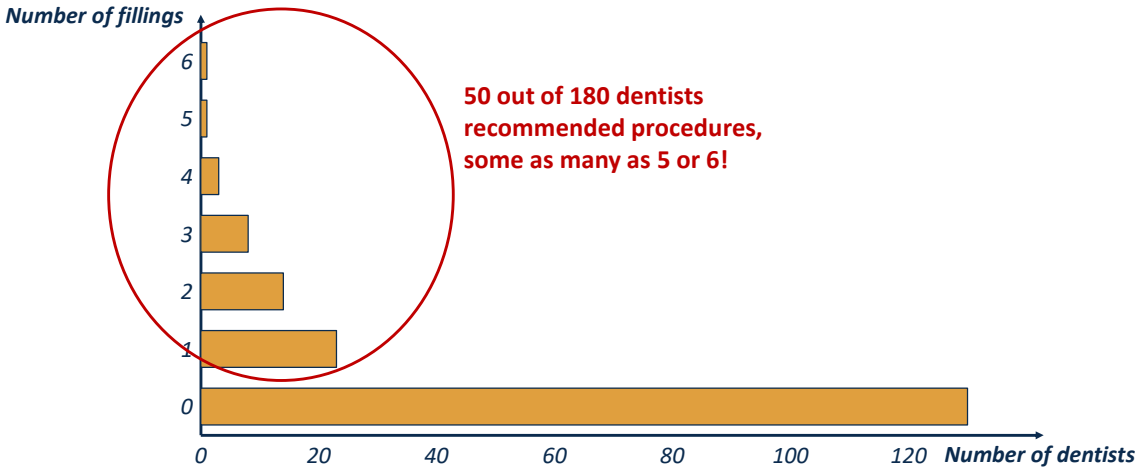
Principal-Agent Relationship: Patient and Dentist

- **Principal:** Patient
- **Agent:** Dentist
- Asymmetric information
- Moral hazard
- Experiment:
  - Send person with perfect teeth to 180 dentists
  - Ask for recommended procedure



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Number of Recommended Fillings by 180 Dentists for a Patient with Verified 'Perfect Teeth'



Source: Gottschalk, F., Mimra, W. & Waibel, C. (2020). "Health Services as Credence Goods: A Field Experiment." The Economic Journal, 130(629), 1346–1383

Solutions to Moral Hazard

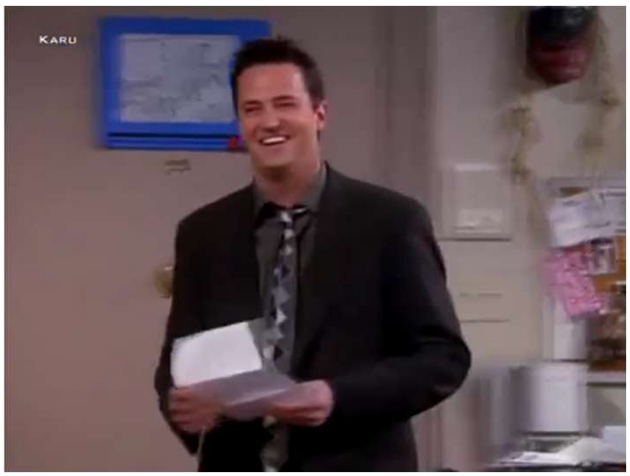
- Address the asymmetric information
  - Collect information: true market value / second opinion → ✗
- Anticipate moral hazard
  - Agents' fees increase by little (3%) if you sell at a higher price → anticipate that the agent will push you to sell → ✗
- Address the incentive misalignment

Accept?
✓ or ✗





Health Insurance and Moral Hazard



Joey Loses His Insurance | Friends Vines, 2014

Source: https://www.youtube.com/watch?v=X8BPfLhH6MA

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Insurance and Moral Hazard

- Health insurance
- Unemployment insurance
- Bank insurance
- Speeding ticket insurance
- "Parental" insurance









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Adverse Selection

- **Adverse selection** can arise when there is **asymmetric information** between buyers and sellers
  - **Used car market:** buyer does not know whether a particular car has engine problems
  - **Labor market:** employer does not know whether a particular worker is productive
  - **Financial market:** a bank does not know if a particular loan applicant is trustworthy
  - **Insurance market:** insurer does not know if a particular individual has high or low risk

Illustrating Adverse Selection in the Used Car Market

- There are two types of cars:
  - "bad" (lemons)
  - "good" (plums)
- The seller knows the type of car
- The buyer does not know the type

Illustration: separate slide deck
under "Lecture Slides" → "Module 8 – Lemons Market"

Result: "good" cars are pushed out of the market → market failure

Adverse Selection Lessons

- Adverse selection leads to **inefficiencies**: untapped business
  - Bad for firms, as they could also profit from low-risk individuals
  - Bad for consumers, because low-risk types do not buy insurance
- Learn to **anticipate** adverse selection. If you ignore it, you likely lose money
- Next: ways to cope with adverse selection

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Ways to Address Adverse Selection

- **Gather information:** e.g., car insurance prices by risk group, to *partly* alleviate adverse selection
- **Screening:** e.g., **screen** auto insurance applicants by driving histories, or let them choose combinations of premia / deductibles
- **Risk pooling: mandate** people to buy insurance so they can't select in or out (e.g., auto insurance, Medicare, Obamacare)
- **Reputation: earned** from repeated interactions (e.g., Yelp, referrals)
- **Signaling**: if a party can credibly **signal** their type, adverse selection (and/or moral hazard) may be reduced or eliminated

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Signaling

- Suppose you could credibly signal to the insurance company that you are a low-risk customer: send a signal
- The problem is that high-risk customers may have an incentive to lie and pretend to be low-risk customers
- For signaling to work, sending the **signal must be costly** to the sender
- The cost must be high enough to keep bad types from wanting to incur it, i.e., from pretending they are the good type



Progressive Snapshot

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Signaling and Education (1)

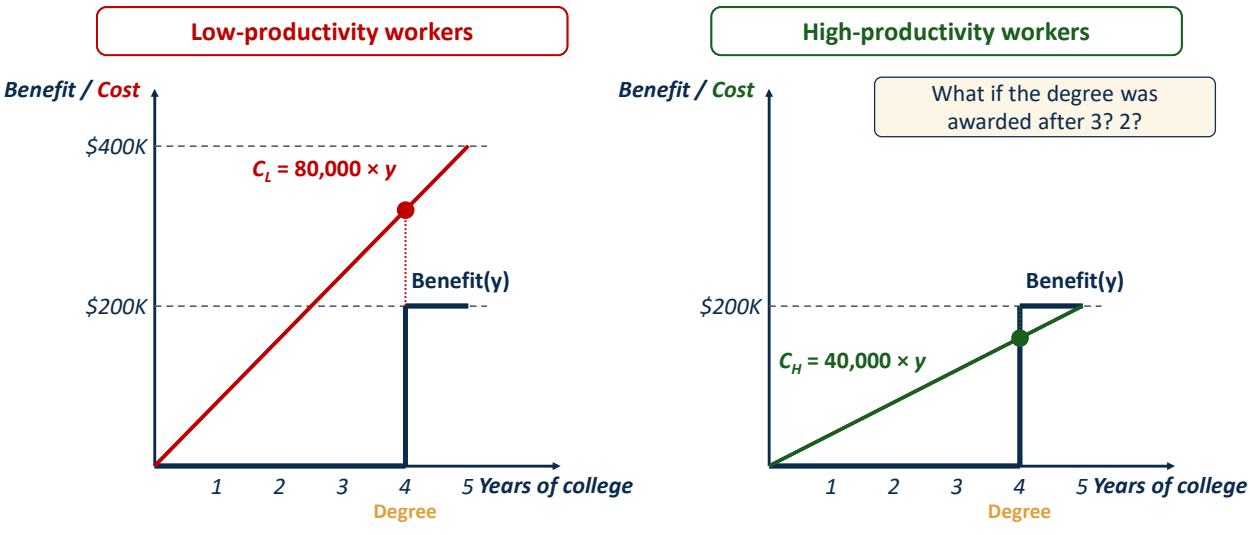
- Split workers half-half:
  - "High type": high productivity (MRPL = $70K per year)
  - "Low type": low productivity (MRPL = $50K per year)
- Each worker is expected to work for 10 years
- The firm cannot tell the type of the workers
- Workers can **signal** their type by investing in education
- Education is assumed **not to affect productivity**!

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Signaling and Education (2)

- The cost per year of college for the high type is:
  - C_H = \$40K × years of education
- The cost per year of college for the low type is:
  - C_L = \$80K × years of education
- Obtain a degree after 4 years
- The firm follows the rule: "Pay \$50K to workers without degree; pay \$70K to workers with degree"
- For simplicity: benefits over 10 years: \$20K × 10 = \$200K
- **What is the equilibrium? Is it separating?**

Signaling and Education (3)



Signaling and Education: Take-Away

- Education can serve as a signal of one's productivity
- The signal must be **costly** to yield separating equilibria
  - i.e., education must be too costly for the low type to pursue it

More Examples for Signaling

- What do King Solomon and David Lee Roth (Van Halen) have in common?
- **Both used signaling!**
- **Medieval ordeals**:
  - Data on 308 cases that entered trial by ordeal in 13th-century Hungary (grab hold of a red-hot iron bar)
  - 100 aborted (defendant confessed)
  - 208 ordeals performed: only 78 were burnt! What happened?
- Last example: 'crazy' claims by scammers



King Solomon



David Lee Roth

Source: Leeson, P. T. (2012). "Ordeals." The Journal of Law and Economics, 55(3), 691–714

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3	**Elective Classes in Economics**	**In class**

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"Global Economics and Business Cycles"

Sebastian Edwards, Brian Wheaton

Core Class

- Analysis of the national and global economy: the environment in which every firm operates
- Financial crises, recessions, and recoveries
- The role of the Federal Reserve in the macroeconomy
- Costs and benefits of a monetary union
- How do countries grow over long periods of time?
- Why do some countries grow faster than others?



Sebastian Edwards



Brian Wheaton

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"Market Games: How to Be Ahead of Competition"

Kathleen Ngangoue

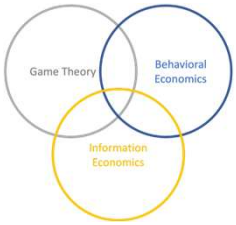
Elective



Kathleen Ngangoue

- What happens in markets when some parties are **less informed** than others?
- How **strategic** do you have to be to conceal, reveal, or infer information?
- Who is paying attention to what? How can we monetize information in today's **Attention Economy**?

This course combines three disciplines: **Game Theory** (how to be strategic), **Behavioral Economics** (which pitfalls to avoid), and **Information Economics** (understand today's trends in the Information Economy)



"Global Trends"

Romain Wacziarg

Elective



Romain Wacziarg

- Explores **recent global trends** affecting business: economic and geopolitical evolutions with first-order effects on management practices and business opportunities worldwide
- A **big-picture class** covering a wide range of topics, e.g.:
  - Is globalization reversing?
  - How is automation altering the economy?
  - How will the rise of new powers and new interest groups affect the conduct of business and policy?
  - What are the major sources of global risk and uncertainty?
- Debate and **varying viewpoints** among participants are encouraged, both in class and on online forums

"International Business Economics"

Jerry Nickelsburg

Elective



Jerry Nickelsburg

- Study the rich fabric of the international economic network
- Analyze the impact of global events (e.g., the war in Ukraine) and how they affect risk and business decisions
- Guest speakers will discuss foreign direct investment, intellectual property, and multinational retail
- The course is centered around a practical student business case of a cross-border opportunity

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"Business Forecasting"

Jerry Nickelsburg

Elective



Jerry Nickelsburg

- Forecasting (the art of constructing forward-looking statements) permeates all aspects of business decision making
- This is a hybrid class that develops a practical tool kit for investment, business development, and business plans
- Guest speakers will discuss forecasting in the airline and finance industries
- Students will apply the techniques in a quarter-long business case forecasting project

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